

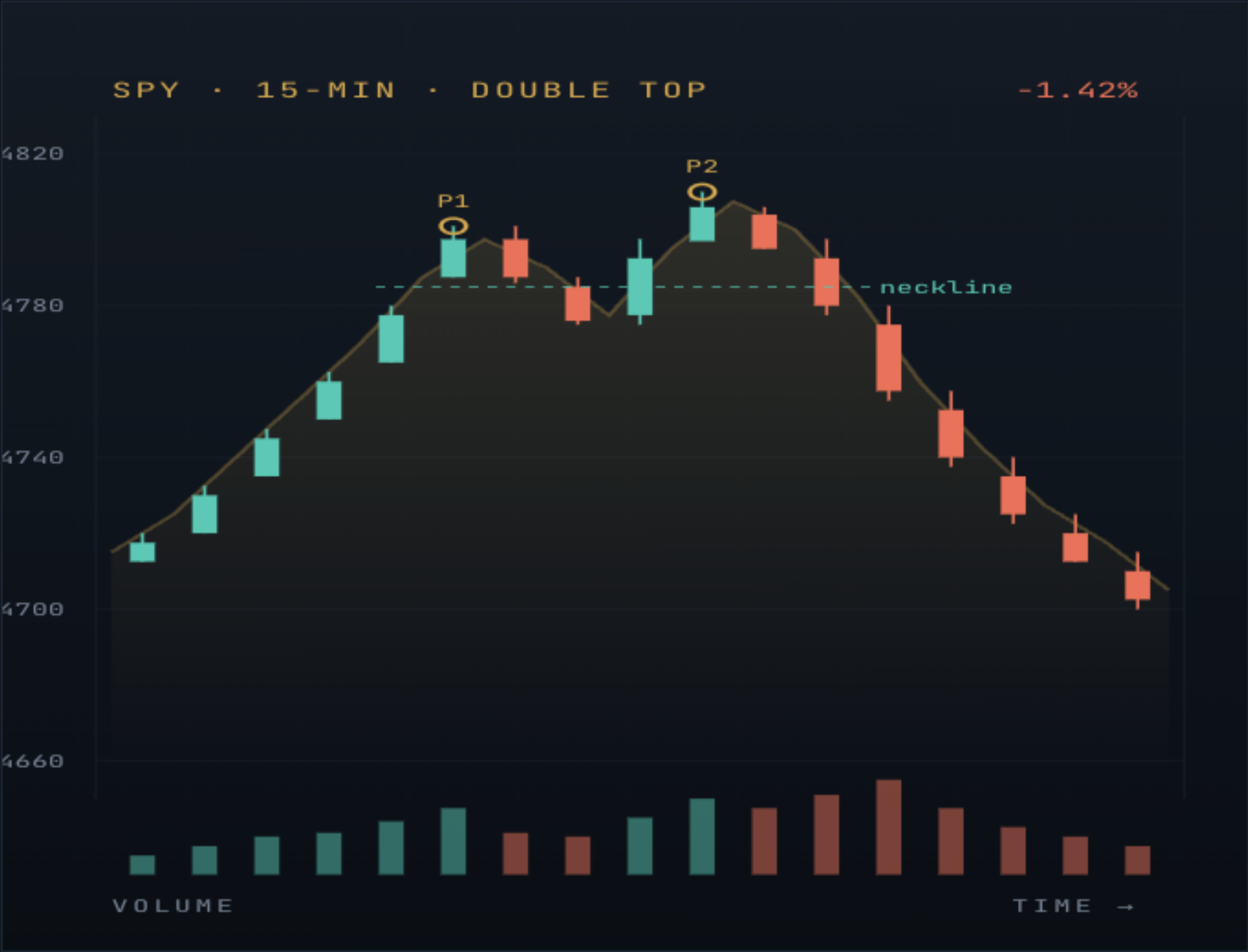
WM9B7 • ARTIFICIAL INTELLIGENCE & DEEP LEARNING
GROUP ASSESSMENT • APRIL 2026

A CONSULTANCY PROPOSAL
FOR INTELLISYS LTD.

GitHub: <https://github.com/oskarone21/Candlestick-Pattern-Recognition>

AUTOMATED CHART PATTERN RECOGNITION

Four reversal patterns. *One pipeline.*



DOMAIN
Finance • Equity Markets

DATASET
SPY 15-min • 2008–2021

PATTERNS
Four reversal patterns

ROLE
Analyst decision-support

What IntelliSys's clients *need.*

*Finance analysts still scan charts by hand. That slows coverage, creates inconsistency, and makes vendor black boxes hard to defend. Our product pitch is an **explainable screening layer** that narrows the review queue without removing human judgement.*

THE PROBLEM • MANUAL TODAY

- **Human-only.** Analysts eyeball charts one at a time.
- **Doesn't scale.** One analyst covers a handful of symbols. Not hundreds.
- **Inconsistent.** Two analysts disagree on the same chart.
- **Black boxes fail audit.** Signal vendors don't survive due diligence.

WHAT INTELLISYS CAN CREDIBLY SELL NOW



Source: Stage-A run `dl_rerun_aug_windowminmax_stageAdata_20260419`. The results slide unpacks where detection quality is strong, where analyst review is still needed, and where PnL stays positive.

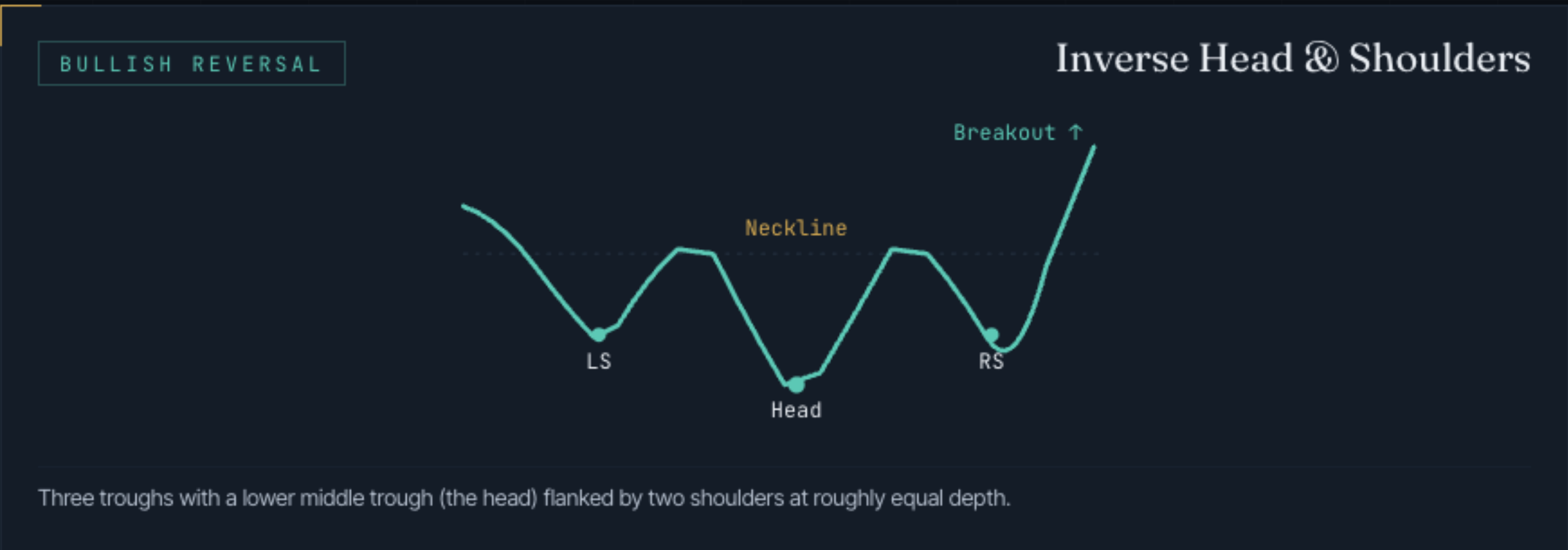
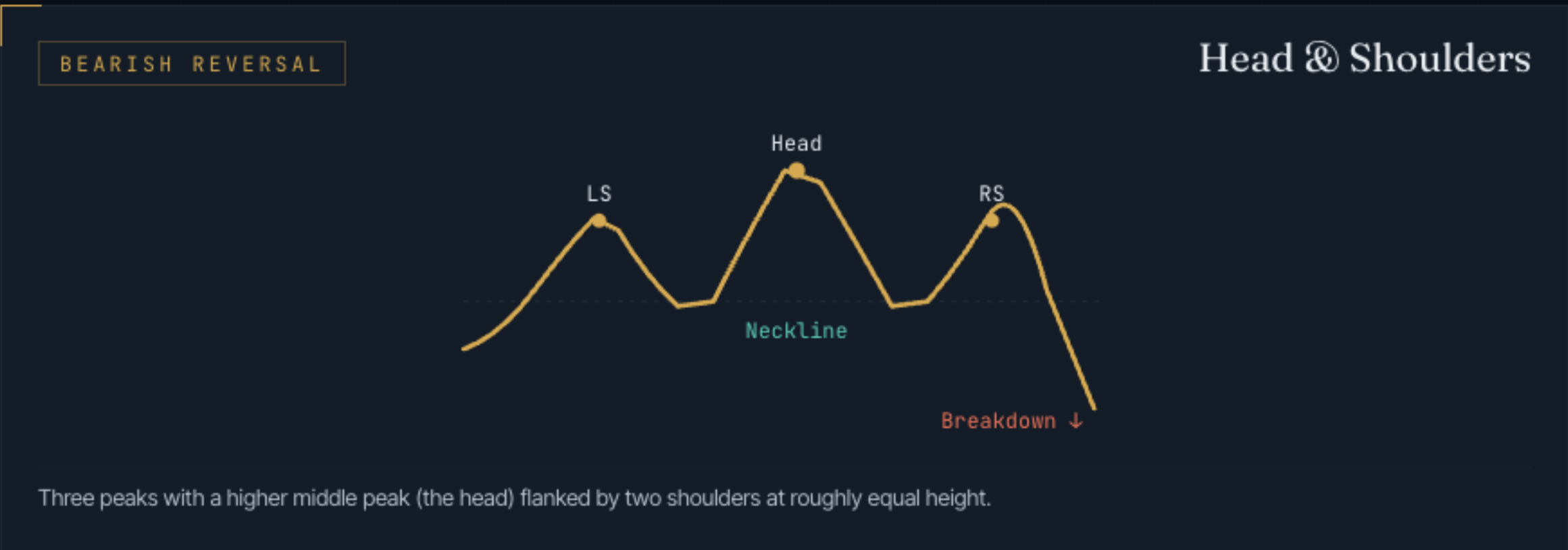
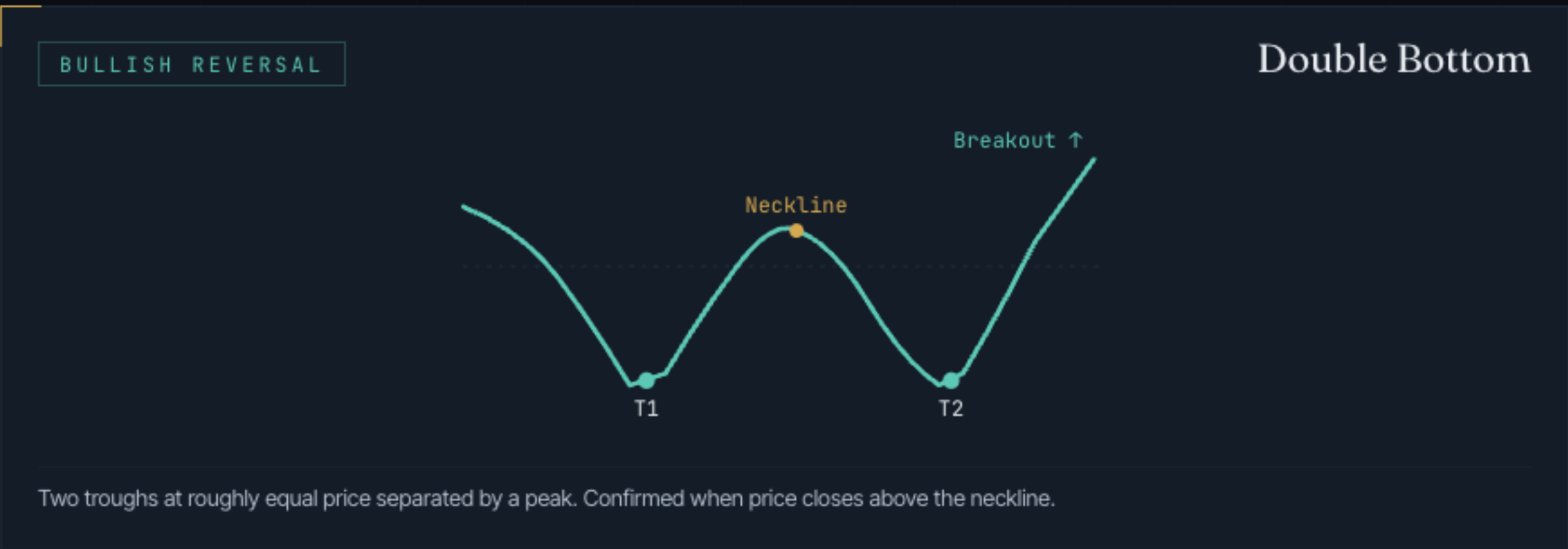
LITERATURE

Cavalcante et al. (2016)
ML in finance — decision-support over autonomous trading. DOI: [10.1016/j.eswa.2016.02.006](https://doi.org/10.1016/j.eswa.2016.02.006)

Lo, Mamaysky & Wang (2000)
Algorithmic pattern detection carries real information. DOI: [10.1111/0022-1082.00265](https://doi.org/10.1111/0022-1082.00265)

What we are *looking for*.

Each pattern is defined by strict geometric rules before any model is trained: peaks, troughs, necklines and breakout conditions. This is what the analyst scans charts for, and what our rule-based labeller encodes as ground truth.



ACADEMIC REFERENCE
FOR DEFINITIONS

Wan & Si (2017) formalise chart-pattern specifications and present them as a reference model for unambiguous definitions in financial time series. That is the literature basis for treating these four shapes as rule-based labels rather than subjective sketches. DOI: 10.1016/j.ins.2017.05.028

S&P 500 ETF *one-minute bars.*

Dataset • SPY *2008–2021*

SPY at 1-minute resolution. The most-liquid US equity ETF — minimal microstructure noise, cleanest geometric patterns, thirteen years covering three major regimes.

2.07M

RAW 1-MIN BARS

86.5k

15-MIN BARS (TRAINED ON)

SOURCE	Kaggle • CC0 1.0 licensed
DAYS	3,339 trading sessions
FIELDS	OHLCV
REGIMES	2008 crisis • 2010s bull • COVID

Regime diversity matters. A single-regime model learns the regime, not the pattern.

Ingestion *∅* *cleansing*

- 01

Session filter
NYSE regular session only • 09:30–16:00 ET.
- 02

Timezone audit
America/Denver → America/New_York. DST-checked.
- 03

Duplicate & NaN removal
Three quality reports written to disk.
- 04

1-min → 15-min resample
Microstructure noise swamps geometry at 1-min. 15-min = ~26 bars/session, enough for multi-peak patterns.
- 05

Nadaraya-Watson smoothing
Causal kernel regression, AICc bandwidth. Keeps structure, cuts noise.

Why deep learning *and which models.*

Why deep learning *here*

- **Sequential structure.** Patterns are defined over time. Classical models lose ordering.
- **Non-linear geometry.** Rules combine bars in non-additive ways. Linear models underfit.
- **Feature learning.** No hand-crafted features — the model learns its own.
- **Scalable.** One architecture generalises from SPY to any instrument.

Operationally, missed opportunities matter too. Because this is a screening workflow, not auto-execution, selection stays **recall-aware** with a **0.50 precision floor** so noisy patterns do not flood the analyst queue.

Models we *trialled*

CHAMPION • ALL 4 PATTERNS IN STAGE-A RUN

TCN

Temporal Convolutional Network. Dilated causal convolutions — wide receptive field, no future leakage, fast training. **Bai et al.** 2018 • [arXiv:1803.01271](#)

BASELINE • SEQUENCE

LSTM

Gated recurrence. Classic financial sequence baseline — tells us whether TCN's extra architecture is earning its keep. **Hochreiter & Schmidhuber 1997**

Trained, split *and tuned.*

Time-ordered split • embargo gaps

Random splitting leaks future into past. In the Stage-A run we split strictly in time order with an **enforced purge gap** between train, validation and test, so the model never learns from future windows by accident.

TRAIN	Earliest SPY bars. Train-only augmentation with 1:3 hard negatives.	60%
EMBARGO	Derived purge gap from lookback + confirmation.	Enforced
VAL	Threshold locking, early stopping, champion selection.	20%
EMBARGO	Same leakage guard before the final holdout.	Enforced
TEST	Most recent bars. Touched once.	20%

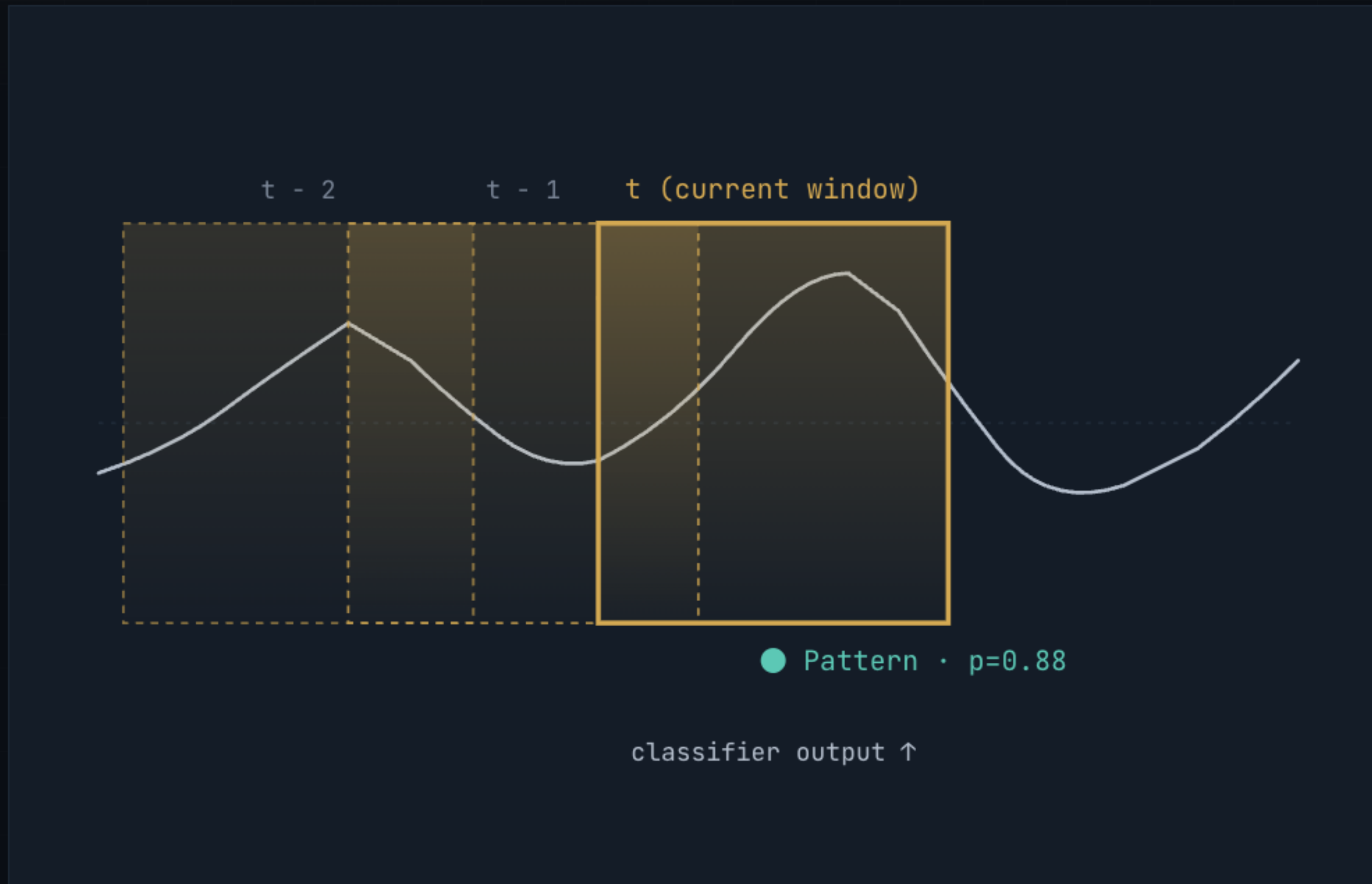
Augmentation on train only — val & test reflect real distribution.

Parameter tuning

EPOCHS	Up to 50, early stop patience 8
OPTIMISER	AdamW • learning rate 0.0003 • weight decay 0.0001
BATCH	64 • reproducible prototype profile
LOSS	Weighted BCE • class-frequency weighting
HP SEARCH	Bayesian/TPE hyperparameter search on validation
THRESHOLD	Per-pattern, locked on val
SELECTION	Recall-aware champion selection with precision floor 0.50
CALIBRATION	Platt scaling on sequence-model probabilities
NORMALISATION	Per-window min-max (price-drift safe)

Sliding windows and *locked thresholds*.

The model does not look at the whole price history at once. It slides a fixed-length window across the series and classifies each window independently.



THE AMBER BOXES

Each box is one sliding window. The configured lookback is 80 bars; this diagram only sketches a few example windows. That span is long enough to capture multi-swing reversal geometry on 15-minute data.

THE GREY LINE

Nadaraya–Watson smoothed price. Both labeller and model see this.

THE TEAL DOT

Positive classification. Fires only above the per-pattern threshold locked on val.

WHY THIS MATTERS

Thousands of windows scanned per day. Analyst only sees what clears threshold.

Detection quality, *then trading value.*

The current prototype shows a more realistic operating profile: precision varies by pattern, and the backtest helps show which alerts are worth escalating toward action.

The headline cards below are a focused performance snapshot for the prototype. The live dashboard's full scoreboard is a broader model-comparison view, so the figures are not intended to match one-for-one.

DOUBLE TOP

TCN

94.7%
PRECISION

F1

87.8%

SUPPORT

22 pos

TEST PNL

+\$11.9k

HEAD & SHOULDERS

TCN

52.6%
PRECISION

F1

69.0%

SUPPORT

40 pos

TEST PNL

-\$45.4k

INVERSE H&S

TCN

60.0%
PRECISION

F1

75.0%

SUPPORT

36 pos

TEST PNL

+\$23.3k

DOUBLE BOTTOM

TCN

95.5%
PRECISION

F1

95.5%

SUPPORT

22 pos

TEST PNL

+\$20.9k

Thresholds are the minimum model-confidence levels required before an alert is shown. We tune them *per pattern* because some shapes are naturally noisier than others, so one cutoff would either miss good candidates or flood the analyst with weak ones.

MACRO SUMMARY

Prototype evaluation snapshot · four patterns covered

81.8%
MACRO F1

75.7%
MACRO PRECISION

94.3%
MACRO RECALL

TRADING QUALITY • SAME EVALUATION SNAPSHOT

PATTERN	TRADES	WIN RATE	TOTAL PNL	PF
Head & Shoulders	65	32.3%	-\$45.4k	0.50
Inverse H&S	55	50.9%	+\$23.3k	1.50
Double Top	19	52.6%	+\$11.9k	1.53
Double Bottom	21	61.9%	+\$20.9k	2.05

Honest limitation: head & shoulders is still the noisy exception. Precision is materially lower and test PnL is negative, so it is best treated as a screening alert, not an action signal.

Three audit layers, *one analyst workflow.*

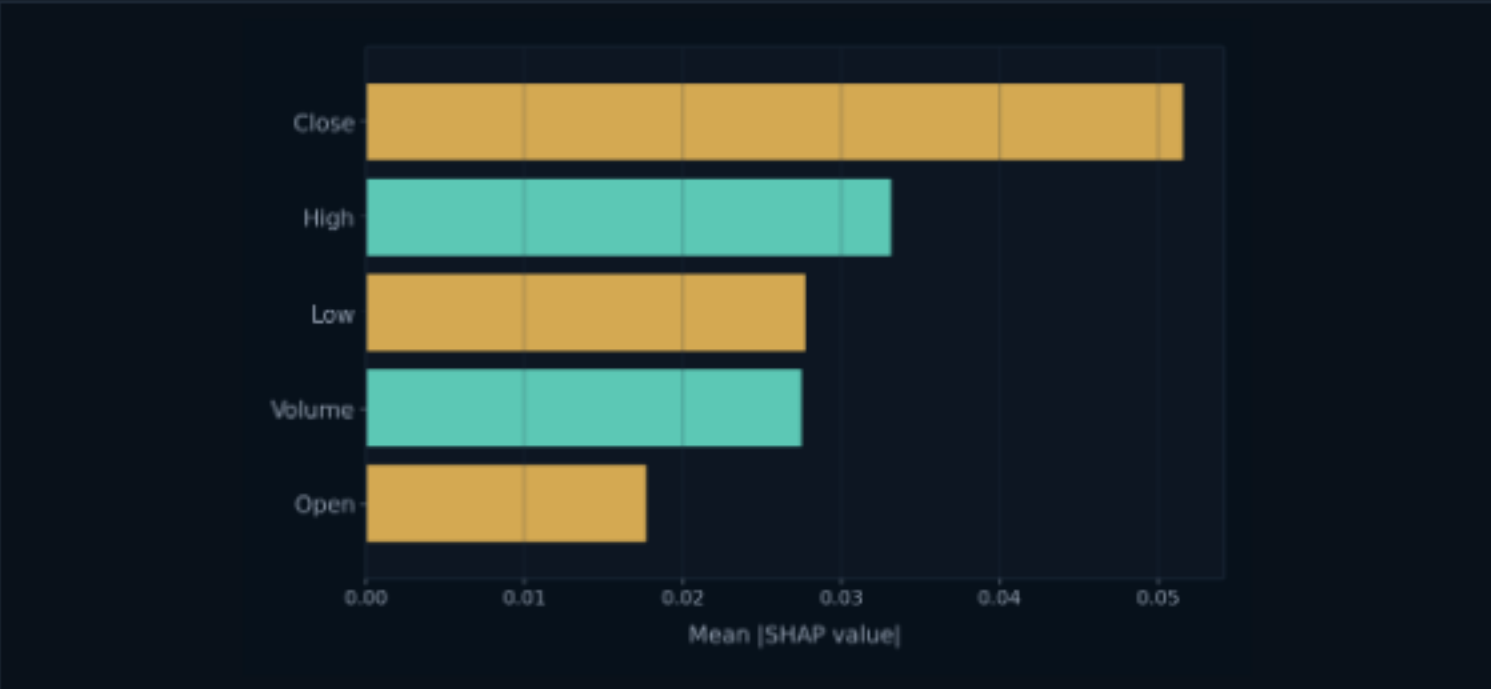
The analyst sees the chart first, the encoded rule set second, and the SHAP attribution third. Below is the **real audit trail** for a Stage-A double-bottom true positive from the run we actually present.

GALLERY • REAL TRUE POSITIVE



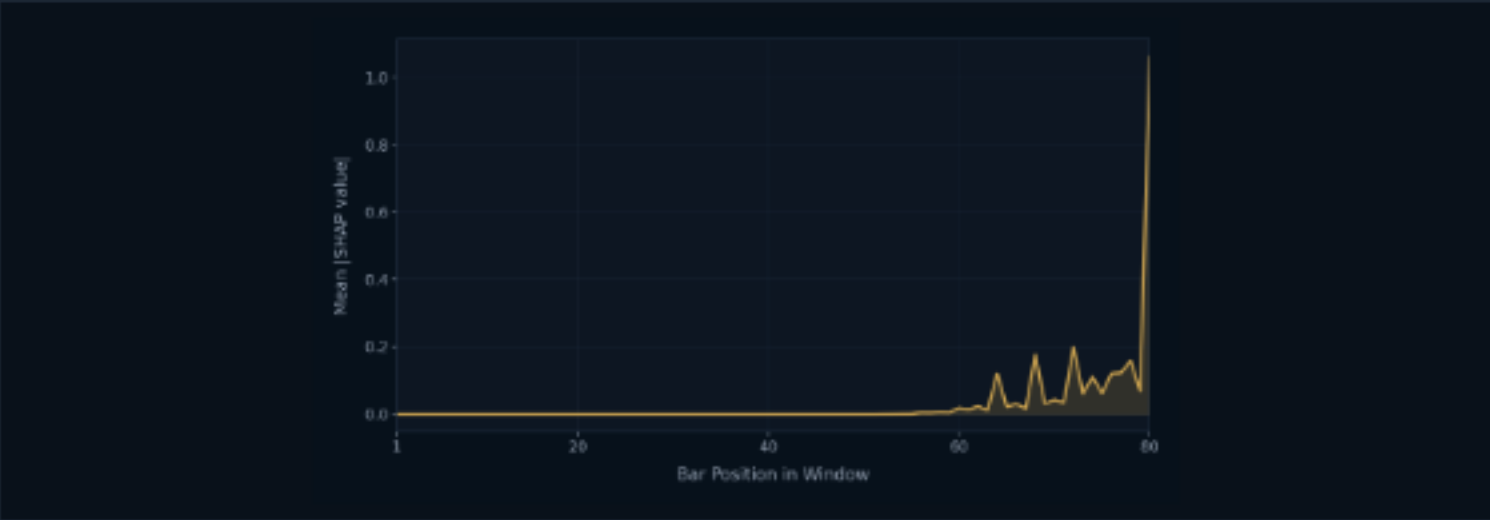
Stage-A gallery counts for double bottom: 12 TP, 1FP, 1FN.

SHAP • GLOBAL FEATURES



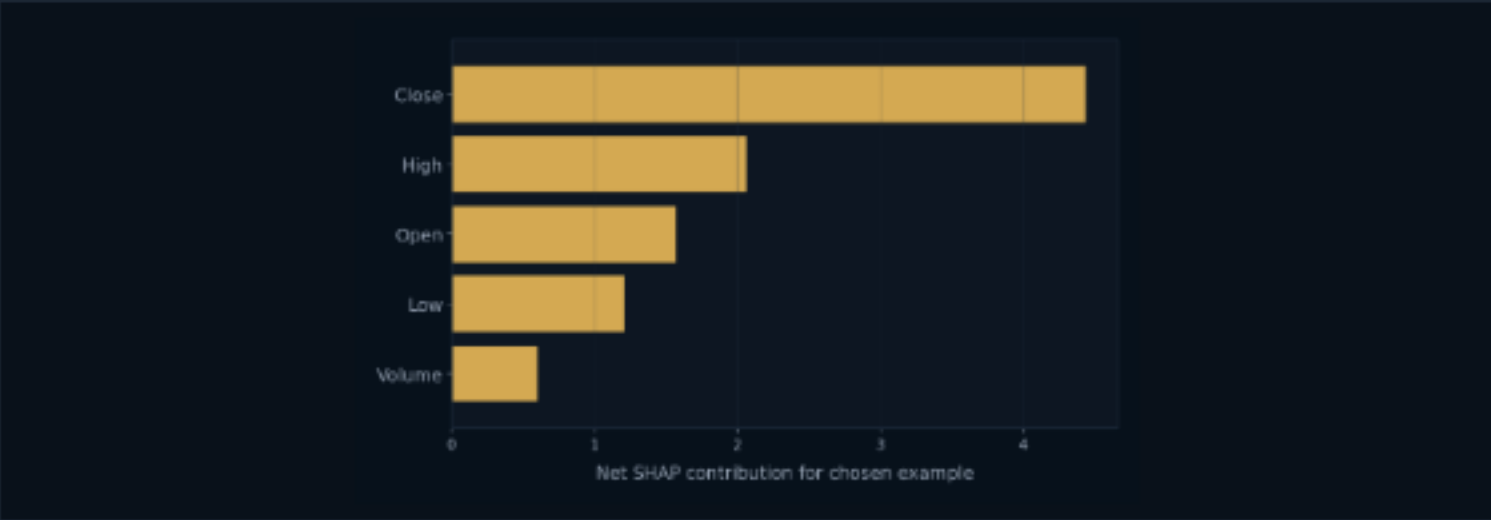
Global ranking is stable: **Close**, **High**, then **Low**.

SHAP • TIME FOCUS



The strongest contribution arrives in the final confirmation bars, which is why the right edge of the 80-bar window matters most.

SHAP • LOCAL DRIVERS



For the chosen true positive, **close** contributes most to the decision, with **high** and **low** reinforcing the same call.

PRIMARY LENS

Gallery view
Every alert becomes a chart the analyst can inspect immediately. This is how we keep human review fast instead of theoretical.

SECOND LENS

Rule trace
The label still ties back to YAML geometry, neckline, breakout and volume rules. Nothing is untraceable.

DEEP AUDIT

Real SHAP outputs
For double bottom, **close**, **high**, and **low** dominate globally, and the signal concentrates late in the 80-bar window.

BUSINESS MEANING

Trust before escalation
When the chart, the rules, and the SHAP pattern agree, the alert becomes a decision candidate rather than just a screener output. That is the product value IntelliSys can defend.

FCA DP22/6 and the EU AI Act push transparency from “nice to have” into a commercial requirement. Gallery images plus rule trace plus SHAP make the system reviewable by both analysts and compliance teams.

What could *go wrong*.

Technical risks *& mitigations*

REGIME	Regime drift after deployment Time-based splits, embargoes, and retraining against fresh periods reduce the chance that the model only memorises one market phase.
SUPPORT	Pattern quality diverges The same pipeline performs differently by pattern. In Stage-A, head & shoulders stays recall-heavy but noisy, so we rank alerts by pattern quality instead of treating every signal equally.
LEAKAGE	Future-bar leakage Time-based splits, enforced purge gaps, causal smoothing and locked validation thresholds keep train/val/test boundaries auditable.
IMBALANCE	Extreme class imbalance Weighted BCE, train-only positive augmentation, 1:3 negative downsampling, calibration and support gating keep rare events usable without leaking into test.

Ethical *principles*

TRANSPARENCY	Every output is explainable Gallery, rule trace and real SHAP artifacts sit behind each alert. That gives analysts and regulators a review path instead of a black box.
ACCOUNTABILITY	Human-in-the-loop Decision-support, not autonomous agent. Analyst owns the call.
FAIRNESS	Bias declared Single-instrument and temporal-regime bias acknowledged, not hidden.
SOCIETAL	Herding & alert fatigue Human review remains mandatory, thresholds stay locked, and the pilot is judged on false-positive control and analyst adoption rather than alert volume alone.

Start with trust, *then scale what earns it.*

Do not automate the trade. **Productize the decision.** The strongest commercial path is a focused analyst workflow first: faster screening, clearer audit trails, and scale only where the signal keeps proving itself.

01

AVAILABLE NOW

A signal analysts can trust

- Four reversal patterns in one auditable workflow
- Strong detection quality with explainability built in
- Three patterns already show positive test PnL
- Every alert can be reviewed through chart, rules and SHAP

02

FIRST 90 DAYS

Start small. Learn fast.

- Pilot one instrument or a tightly chosen watchlist
- Put ranked alerts in front of analysts, not execution engines
- Prioritise by detection quality, support and trading value
- Measure adoption, trust and false-positive control from day one

03

AFTER PROOF

Scale only what earns the right

- Expand only where the signal stays strong under review
- Retune geometry per instrument before rollout
- Add drift monitoring and versioned explainability packets
- Keep the product disciplined: screening first, execution never by default

Live-data headroom: the current evidence uses regular-session historical bars only. Broader, cleaner live feeds may improve usefulness or coverage, but that is a pilot hypothesis to test, not a proven gain claim.

CURRENT EVIDENCE	Strong screening quality • mixed monetisation by pattern	DECISION	Analyst-first workflow	INFRA	Auditable, explainable, deployment-ready	ASK	Approve a focused 90-day pilot
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What this *is built on.*

Every technical decision on the previous slides traces back to one of these. DOIs given so claims can be verified directly.

Pattern detection @ financial ML

[1] Lo, A.W., Mamaysky, H. & Wang, J. (2000). Foundations of Technical Analysis. *Journal of Finance*, 55(4). DOI: 10.1111/0022-1082.00265

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Deep learning architectures @ XAI

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Also referenced: EU AI Act Regulation (EU) 2024/1689 • FCA DP22/6 (2022) *AI and Machine Learning in UK Financial Services* • SEC Risk Alert (2024) on AI governance • Kaggle dataset: [gratefuldata/intraday-stock-data-1-min-sp-500-200821](#) (CC0 1.0 universal licence).

Tools used. *Ownership intact.*

*In compliance with Warwick University Regulation 11 and WMG AI guidance. This project operates under the *AI Collaboration* scale as specified in the assessment brief.*

Tools used

CLAUDE • ANTHROPIC	Code debugging • error explanation • technical write-up refinement.
AI SCALE	AI Collaboration (per WM9B7 brief).

Our commitment

All AI-generated content was critically evaluated and modified. Original coding, design and implementation decisions are our own. Outputs were adapted and extended to demonstrate genuine ownership.